SpeechLP

# SpeechLP®

## Research Foundation & Logic Model

ESSA Tier 4 “Demonstrates a Rationale” Evidence Packet

**ESSA TIER 4**

Demonstrates a Rationale

THIRD-PARTY REVIEW PENDING


**Prepared by:**

Mari Gottlieb, M.A., CCC-SLP
Cofounder and Co-CEO

Alissa Fox, M.S., CCC-SLP
Cofounder and Co-CEO

Shail Silver
Cofounder and Chairman



NOVEMBER 2026

**ESSA TIER 4**

Demonstrates a Rationale

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# DEMONSTRATES A RATIONALE

ESSA Tier 4 Evidence Statement for SpeechLP

SpeechLP

This packet documents how SpeechLP's design is grounded in published research on speech sound development and intervention, describes the program's features and intended outcomes in a logic model, and outlines a planned study of its effects. Together these components address the Every Student Succeeds Act (ESSA) criteria for Tier 4, "**Demonstrates a Rationale**."

*Independent third-party review pending.*

| | |
|---|---|
| **Reviewer name, credentials**<br>Organization | **DATE** |

# Executive Summary

**SpeechLP is a mobile and web app built by practicing speech-language pathologists, that gives children frequent, structured, and measured practice of speech sounds at home and at school. Through its companion SLP tool, SpeechTherapist.app, speech-language pathologists assign and monitor practice dosage and track outcomes over time to help drive treatment decisions.**

## What SpeechLP does

A 15-word-per-level screener for ages 3–9, aligned with consonant acquisition norms (McLeod & Crowe, 2018; Crowe & McLeod, 2020), identifies children's most difficult sounds. In short video lessons, two certified SLPs model each sound and the child practices saying it. Speaking games draw on more than 1,500 target words across all 24 English consonants. Automated speech recognition gives feedback on every word, and progress is reported to parents, teachers, and SLPs.

## Who it serves

Children ages 2–9 who are working on speech sounds, whether or not they have a diagnosed speech sound disorder. Many families use it alongside therapy; others use it before they have an SLP or while waiting for services. It does not replace an SLP evaluation. Ages are entered by families, so the initial study also included a few older children.

## Rationale

Speech sound disorders are common, affecting an estimated 2.1%–23% of children aged 4–6. Children whose speech difficulties persist, or occur alongside language difficulties, are at greater risk for later reading and spelling problems. Treatment is effective, and research consistently points to practice amount, frequency, and feedback as drivers of progress. Yet therapy time is limited and home practice is rarely measured. SpeechLP is designed to increase practice and make it visible.

## Initial evidence

A descriptive retrospective study of 101 returning children found that app-scored accuracy on the same sounds rose from 69.4% to 74.9% between two consecutive months (+5.5 percentage points, 95% CI 0.4 to 10.4). The scoring is not yet validated against SLP judgment, and there was no comparison group, so this is not evidence of effectiveness. It is a starting point for further studies.

## ESSA status

This packet documents a research-based rationale and logic model and a completed initial study, with prospective evaluation planned, addressing the requirements for **ESSA Tier 4, "Demonstrates a Rationale."** Independent third-party review is pending.

## Evidence goals

1. **Validate scoring** against several SLP raters, acoustic measures, and parent judgments.
2. **Study home practice dose** using de-identified existing data.
3. **Test effectiveness** in prospective studies with a comparison group, building toward Tier 3 or Tier 2 evidence.

| ~15 | 1,500+ | 2 | +5.5 pp |
|---|---|---|---|
| screener words per age level | target words, all 24 consonants | CCC-SLPs modeling every sound | app-scored accuracy, month 1 to 2 (n = 101) |

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# Understanding ESSA Evidence

**Educators search for high-quality, evidence-based interventions to strengthen grant applications, to support comprehensive and targeted schools, and to implement new programming. Evidence requirements under the Every Student Succeeds Act (ESSA) are designed to help states, districts, and schools identify programs, practices, products, and policies that work across various populations.**

Educational programs document their evidence of design, effectiveness, and impact in order to be eligible for federal funding. ESSA defines four levels of evidence: strong (Tier 1), moderate (Tier 2), and promising (Tier 3) evidence of effectiveness, or a program that demonstrates a rationale to be effective (Tier 4). Under ESSA, Tier 4 ("demonstrates a rationale") describes an activity with a well-specified logic model informed by research or evaluation, accompanied by ongoing efforts to examine its effects (ESEA §8101(21)(A)(ii); U.S. Department of Education, 2016).

**This packet addresses the requirements for Tier 4:**

- ✓ Documentation of how the product's design relates to intended outcomes, with corresponding published academic research
- ✓ A logic model describing the product's resources, activities, outputs, and outcomes
- ✓ An initial descriptive study has been completed (Silver et al., 2026), and a prospective evaluation is planned (see "Planned Evaluation")
- ★ A third-party research organization reviews the documentation for ESSA validation (pending)

**ESSA TIER 4**

Demonstrates a Rationale

THIRD-PARTY REVIEW PENDING

When speech-language tools are built on what is known about how children acquire and practice speech sounds, families, teachers, and SLPs can extend high-quality practice well beyond the therapy room. This packet maps each SpeechLP feature to that research and lays out how its effects will be measured.

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## SpeechLP Supports Speech Sound Development



SpeechLP is a mobile and web app that helps children practice clear speech sounds between, or in addition to, sessions with a speech-language pathologist. A brief age-based screener of about 15 words per age level identifies which sounds are tricky for a child. The app offers short SLP-led video lessons and a guided learning path of playful practice games drawn from a library of more than 1,500 target words. SLPs use the companion **SpeechTherapist.app** to assign practice to students and see their progress.

### A Screener to Focus Practice

Children start with a short try round to get comfortable. A screener matched to their age, from 3 to 9, is built into the early levels of the learning path (from level 2 through level 6 or 9, depending on age) and uses about 15 words to sample the speech sounds expected by that age in initial, medial, and final word positions, so practice can focus on the sounds that need attention rather than on sounds already mastered.



### What Parents Say

C

"She is able to navigate the app independently, which is amazing for her age, and the activities keep her engaged and interested. She actually enjoys practicing, so it doesn't feel like therapy—it feels like play!"
**Carrie G.**
SpeechLP parent (App Store review)

"I honestly didn't think I'd find a speech therapy app my daughter would actually like, but this has been a total surprise. She really enjoys practicing her sounds with it, and we've already seen so much improvement."
**Sheddy4U**
SpeechLP parent (App Store review)

P

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SpeechLP

# SpeechLP® Foundational Research Summary

## Introduction

Speech sound disorders are among the most common communication difficulties of childhood. In a large population-based sample, Shriberg, Tomblin, and McSweeny (1999) found that about 3.8% of 6-year-olds had a speech delay. ASHA reports estimated speech sound disorder prevalence of 2.1%–23% at ages 4–6 and persistent speech sound disorder in about 3.6% of children at age 8 (ASHA, n.d.). Children with speech sound disorders are at increased risk for later literacy difficulties, especially when speech difficulties persist or co-occur with language impairment (Raitano et al., 2004; Peterson et al., 2009; Hayiou-Thomas et al., 2017), and parents of school-aged children with residual speech errors report social, emotional, and academic impacts (Hitchcock et al., 2015). A 2025 meta-analysis found moderate reading deficits in children with speech sound difficulties, both concurrently ($g$ = −0.58) and over time ($g$ = −0.50), though the authors note possible publication bias (Walquist-Sørli et al., 2025).

Effective speech intervention depends heavily on practice. Children need many accurate productions of a target sound, distributed over time, to make lasting gains (Maas et al., 2008; Williams, 2012). Yet therapy time is limited, and practice between sessions depends on families: in a survey of 288 SLPs, 95.8% agreed or strongly agreed that family involvement is essential for intervention to be effective (Sugden et al., 2018). Treatment itself is effective: a meta-analysis of controlled studies with children aged 3–18 found medium-to-high effects on phonological accuracy (Kunnari et al., 2024). The challenge is giving children enough of it.

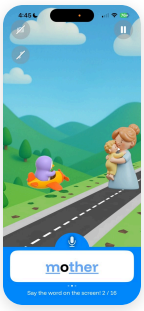
SpeechLP is a mobile and web app that extends speech sound practice beyond the therapy session. It combines a brief screener, SLP-led video lessons, a large library of target words, and game-based word practice, with progress reports for parents, teachers, and SLPs. SpeechLP is designed to supplement, not replace, assessment and intervention by a licensed speech-language pathologist.

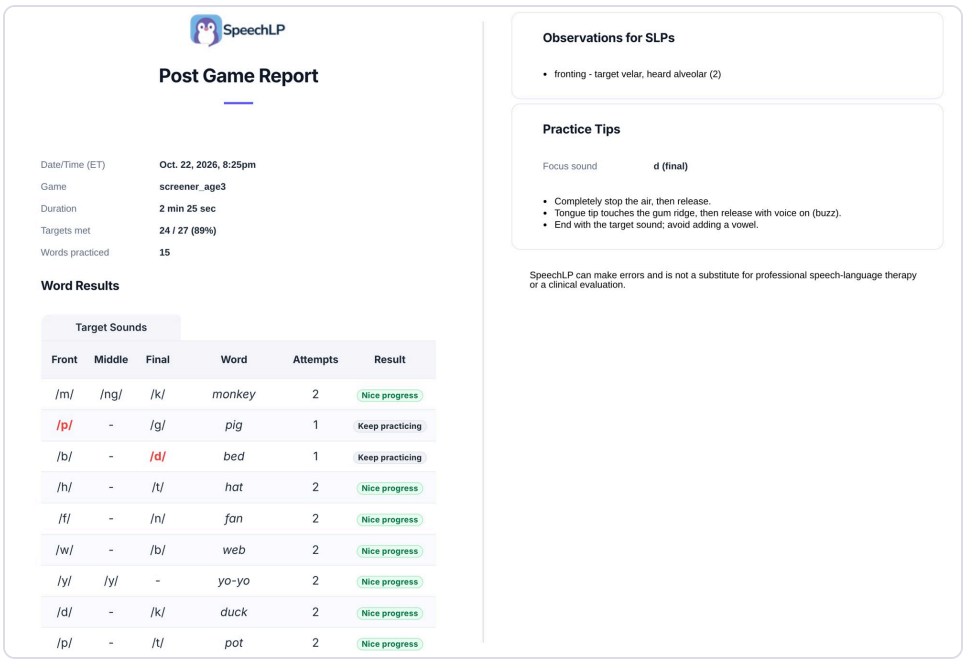
## Research on SpeechLP's Key Differentiating Features

### 1. Age-Based Screening of Developmentally Expected Sounds

Children acquire English consonants in a broadly predictable developmental sequence (Smit et al., 1990). A review of 15 U.S. studies covering 18,907 children found that most English consonants were acquired by age 5;0 under a 90% acquisition criterion, with /ɹ/, /ð/, and /ʒ/ acquired by 5;11 and /θ/ by 6;11 (Crowe & McLeod, 2020). These population norms are not individual diagnostic deadlines, but they indicate which sounds a child of a given age would typically be expected to produce. Identifying which sounds are in error is the starting point for selecting intervention targets (McLeod & Baker, 2017).

SpeechLP has screeners for each age from 3 to 9, each about 15 words long, that sample the speech sounds children are expected to produce by that age. Ages 3, 4, and 5 each have their own set of sounds, reflecting the period when consonants are still being acquired; ages 6 through 9 share one set, since most English consonants are acquired by about age 5 (Crowe & McLeod, 2020). Each screener samples sounds in initial, medial, and final word positions. Words are aligned with consonant acquisition norms (McLeod & Crowe, 2018; Crowe & McLeod, 2020); for example, the age-3 screener targets exactly the 13 consonants that US children acquire by 3;11. Words were selected by SpeechLP's cofounding SLPs, Alissa Fox, M.S., CCC-SLP, and Mari Gottlieb, M.A., CCC-SLP, with the goal of using higher-frequency words that young children are likely to know and that are easy to depict as pictured objects, so that the screener measures speech production rather than vocabulary. Results identify the specific sounds that are difficult for each child, so practice can focus on them. After each screener, parents receive an emailed report with results for each word and sound position, patterns flagged for SLPs, and practice tips, which they can share with their child's SLP or teacher to support earlier conversations about whether a full evaluation is warranted. The screener is not a diagnostic assessment and does not replace an evaluation by a licensed SLP.



SpeechLP

**Post Game Report**

| | |
|---|---|
| Date/Time (ET) | Oct. 22, 2026, 8:25pm |
| Game | screener_age3 |
| Duration | 2 min 25 sec |
| Targets met | 24 / 27 (89%) |
| Words practiced | 15 |

**Word Results**

Target Sounds

| Front | Middle | Final | Word | Attempts | Result |
|---|---|---|---|---|---|
| /m/ | /ng/ | /k/ | *monkey* | 2 | Nice progress |
| /p/ | - | /g/ | *pig* | 1 | Keep practicing |
| /b/ | - | /d/ | *bed* | 1 | Keep practicing |
| /h/ | - | /t/ | *hat* | 2 | Nice progress |
| /f/ | - | /n/ | *fan* | 2 | Nice progress |
| /w/ | - | /b/ | *web* | 2 | Nice progress |
| /y/ | /y/ | - | *yo-yo* | 2 | Nice progress |
| /d/ | - | /k/ | *duck* | 2 | Nice progress |
| /p/ | - | /t/ | *pot* | 2 | Nice progress |

**Observations for SLPs**

- fronting - target velar, heard alveolar (2)

**Practice Tips**

Focus sound **d (final)**

- Completely stop the air, then release.
- Tongue tip touches the gum ridge, then release with voice on (buzz).
- End with the target sound; avoid adding a vowel.

SpeechLP can make errors and is not a substitute for professional speech-language therapy or a clinical evaluation.

*Left: a screener card (word 2 of 16). Right: excerpts from the emailed report parents receive after an age-3 screener, showing word results by sound position, a pattern flagged for SLPs (fronting), and practice tips. Sample report; identifiers removed.*

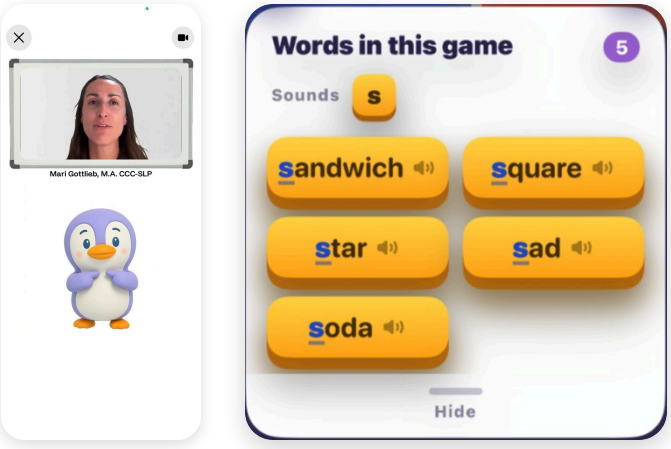
## 2. Expert Modeling Before Every Word

Learning a new speech movement benefits from clear instruction and a model to imitate. Motor learning research applied to speech emphasizes pre-practice—explaining and demonstrating how a sound is made—before a child begins repeated practice (Maas et al., 2008). Video modeling also meets criteria for evidence-based practice in related populations, such as children and adolescents with autism spectrum disorder (Bellini & Akullian, 2007).

SpeechLP's brief video lessons show a certified speech-language pathologist explaining and demonstrating each target sound. The videos feature Mari Gottlieb, M.A., CCC-SLP, and Alissa Fox, M.S., CCC-SLP, who both provide models; Mari delivers the instructional cueing that guides children's productions, which Alissa wrote. During each lesson, the child practices producing the target sound on its own, imitating the SLP's model. Families can open the video sound lessons at any time from the app's Settings. Because lessons are recorded, every child receives a consistent, expert model, and families without easy access to an SLP still see correct placement and production.

**Previewing the words.** Before playing, a "Words in this game" panel shows the target sound and each word the child will practice, with the target sound highlighted and a speaker button to hear the word, so children and parents can preview and listen to every word first.

**A spoken model on every card.** During games, each practice card shows a pictured target word and automatically plays a spoken model of it, which the child can replay as often as needed before speaking. Together, the video lessons, word previews, and card models mean children hear a correct production before every attempt, consistent with research on preparing a learner before practice (Maas et al., 2008).

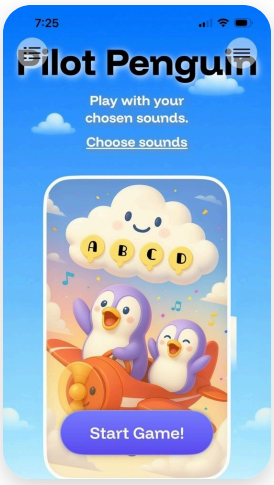



*Left: a video sound lesson, in which a certified SLP models the target sound. Right: the "Words in this game" preview, with the target sound highlighted and a button to hear each word.*

### 3. High-Dose Practice From a Library of 1,500+ Words

The number of practice trials matters. In a small study of two children with childhood apraxia of speech, targets practiced at higher production frequency were learned more quickly (Edeal & Gildersleeve-Neumann, 2011), and a systematic review found that higher treatment intensity tended to yield better outcomes, although the evidence base is small (Kaipa & Peterson, 2016). How practice is distributed also matters: in a randomized study of preschool children receiving the multiple oppositions approach, treatment three times weekly produced greater phonological gains than treatment once weekly when cumulative intervention intensity was held constant (Allen, 2013), and in children with childhood apraxia of speech, twice-weekly motor speech treatment for 10 weeks outperformed once-weekly treatment on articulation and functional communication (Namasivayam et al., 2015). Motor learning research adds that practice amount, distribution, variability, and feedback schedule may affect retention and transfer (Maas et al., 2008).

SpeechLP draws from a library of more than 1,500 words, organized by sound and word position and spanning ages 2–13 (the app is designed for ages 2–9), so children can produce many different exemplars of their target sound rather than repeating the same small set. Practice in SpeechLP currently takes place at the sound level, in the video lessons, and at the word level, in the games. Treatment commonly progresses from establishing a sound to using it in more complex contexts and in conversation (ASHA, n.d.; McLeod & Baker, 2017), and Speech Motor Chaining applies motor learning principles through structured practice that builds from syllables to longer, self-generated utterances (Preston et al., 2019).




*Children choose their target sounds, then practice them across many different words (Pilot Penguin).*

## 4. Motivating, Purposeful Game-Based Practice

Repetitive drill can be hard for young children to sustain. A review of empirical studies found that gamification generally produces positive effects on engagement and motivation, with outcomes depending on the design and context (Hamari et al., 2014). A longitudinal home-practice study of a tablet-based speech therapy game reported significant production gains and increased engagement (Hair et al., 2021), and a systematic review of computer-based speech therapy found positive clinical changes for some children (Furlong et al., 2017).

In SpeechLP, every game is built around speaking. In *Beary Hungry* the child feeds the bear by saying the word; in *Baby Talk* and *Chicken Little* the child must "say it right to make it pass." Children can choose which sounds to play with, progress along a visual map with numbered levels and rewards, and celebrate success with animated characters. These elements are designed to increase the number of productions a child is willing to attempt in a session.




*Every game is played by saying the target words aloud; correct productions earn stars.*

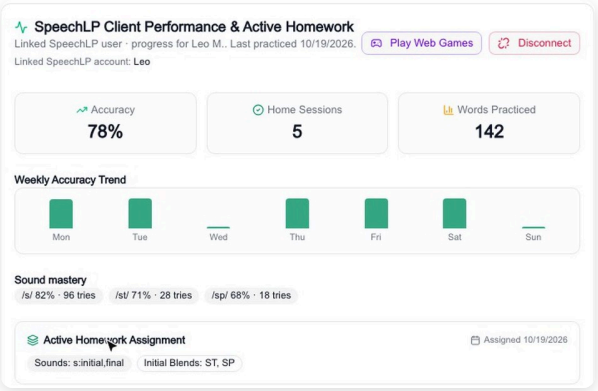
**Automated feedback on every word.** After the child hears the word modeled (see feature 2), automated speech recognition evaluates the child's production against the card's target sound (Silver et al., 2026). Correct productions earn stars; if a word is missed, it times out and the game moves on. Automated feedback in structured speech practice is an active research area. In a single-case experiment, listener-rated /ɹ/ accuracy in practiced utterances was higher immediately after 30 minutes of AI-assisted Speech Motor Chaining practice than before it (Benway & Preston, 2024), and automated tools for analyzing children's speech are a growing field of study (McKechnie et al., 2018). In Swedish children aged 4–10, a speech recognition model rated single words within about 5 percentage points of an expert's own re-ratings on accuracy (Getman et al., 2022), and automated assessment has shown promise in Korean and Mandarin-speaking children (Kim et al., 2025; Lin et al., 2024); none has yet been validated for English home practice. In a single-case study of a self-supervised home speech app, two of four children showed slight improvement on trained words, without improvement on untrained words (Strömbergsson et al., 2026). SpeechLP's acceptance model has not yet been validated against SLP ratings; that validation is the first planned study, and SpeechLP is working to fine-tune the model, with research results guiding where it needs to improve. One known gap: the current model recognizes /ʒ/ (as in "measure") as /tʃ/ ("ch"), so those roughly 29 cards rely on workarounds. SpeechLP never receives any audio.

## 5. Connecting Home, School, and the SLP Through Assignment and Progress Data

Parents and caregivers can play a meaningful role in speech intervention. A review of parent involvement in intervention for speech sound disorders found that parents are frequently involved in delivering home practice and that such involvement can support outcomes (Sugden et al., 2016). In related areas of communication, parent-implemented interventions have shown positive effects on children's language skills (Roberts & Kaiser, 2011). ASHA's national outcomes data found that three times as many children with severe speech sound disorders made significant functional progress in intelligibility when therapy combined 21–30 sessions, scheduled twice weekly, with an SLP-directed home program, compared with children who received fewer sessions or whose plan lacked any of these elements (ASHA, 2024). Yet home practice is rarely measured: surveys of SLPs have not quantified how often it is assigned (Sugden et al., 2018; Tambyraja, 2020), and intervention studies seldom report what it involved (Sugden et al., 2016).

SpeechLP produces reports designed to be read by parents, teachers, and SLPs. Reports summarize screener results and the child's practice and progress over time, giving the SLP information about home practice that would otherwise be unknown, and giving families clear, concrete next steps. Because the app is guided by SLP lessons, parents and teachers do not need clinical training to support practice.

For SLPs, SpeechLP connects to **SpeechTherapist.app**, a companion web app built for the SLP workflow. SLPs invite a caregiver, assign target sounds, word positions, and blends for home practice, and track home sessions, words practiced, and accuracy overall and by sound. Every scored practice item is logged with the word attempted, the sounds recognized, whether it was accepted, and a timestamp, so practice outside the therapy room is measured rather than assumed, and the SLP can adjust instruction based on data.




*SpeechTherapist.app progress view (demo account): accuracy, home sessions, words practiced, weekly trend, sound-by-sound accuracy, and the active assignment.*



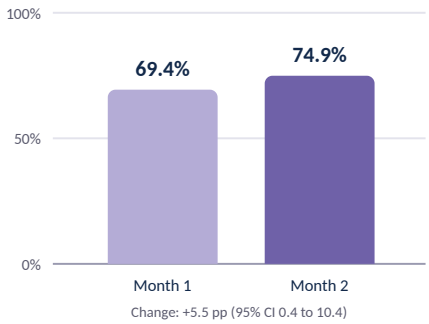
*Children move through a guided learning path and celebrate their progress.*

# Initial Evidence: A Descriptive Retrospective Study

SpeechLP published a retrospective observational study of routine practice records from January 22 through October 7, 2026 (Silver et al., 2026). All authors are affiliated with SpeechLP. Shail Silver, Cofounder and Chairman, performed the initial analysis, with feedback from Jabali Ndungu, Head of Engineering, and Matthew Siu, Software Engineer, who also reviewed the data analysis, statistical methods, and numerical results with him. Mari Gottlieb, M.A., CCC-SLP, and Alissa Fox, M.S., CCC-SLP, Cofounders and Co-CEOs, reviewed the clinical terminology, interpretation, and relevance to speech-language pathology practice. The study followed 101 returning child profiles (55 recorded as ages 2–5 and 33 as ages 6–8) across two consecutive 30-day periods and compared accuracy on identical sound targets within each profile.

## What the study found

- Across 660 matched profile-and-sound pairs (11,902 practice cards), mean accepted-card accuracy rose from **69.4% to 74.9%**, a change of **+5.5 percentage points (95% CI 0.4 to 10.4)**.
- 54 of 101 profiles (53.5%) had higher second-month averages, 34 (33.7%) lower, and 13 (12.9%) unchanged; of 660 individual sound comparisons, 278 rose, 177 fell, and 205 were unchanged.
- Sixteen of 21 reportable sound means were higher in Month 2, including /k/ (+14.4 pp), /z/ (+16.0 pp), and /t/ (+11.1 pp); five were lower, including /ɹ/ (−3.3 pp).
- Practice declined in Month 2: mean sessions per profile fell from 6.73 to 3.44 and mean practice days from 4.70 to 2.71.




Mean accepted-card accuracy on matched sounds, 101 profiles (each profile weighted equally). Source: Silver et al. (2026).

## How to interpret these results

These findings are preliminary and descriptive, and the study's authors state their limits directly. Accuracy reflects the app's automated speech recognition decisions, which have not been validated against SLP ratings, and scoring changed over the study period without adjustment for app version. A card counted as accepted if any attempt succeeded, so scores can reflect repeated tries and listening to the model. The cohort was selected for returning use, there was no comparison group, and many children were in an age range of active sound acquisition, so the change cannot be attributed to SpeechLP practice or interpreted as clinical articulation improvement. The study did not test untrained words or connected speech. Under ESSA, this study does not meet the requirements for Tier 3 (promising) evidence; it is reported here as part of the rationale and as the reference point for the prospective evaluation that follows.

Full manuscript and aggregate data (CC BY 4.0): speechlp.com/research/articulation-practice-two-months · https://doi.org/10.5281/zenodo.23932485

## Planned Evaluation

SpeechLP is committed to building evidence beyond Tier 4 and is inviting university research partners to two starting studies. In each, SpeechLP provides practice data, the app platform, engineering support, and recruitment through connected SLPs; the partner leads study design, ethics approval, and analysis. Scoring is validated against SLPs first; the validated platform is then used to study practice and progress. Validation results will also guide fine-tuning of the app's model. A planned upgrade aims to recognize all 24 consonants, rely less on Apple's speech recognition, and detect phonological processes.

| | |
|---|---|
| **1. Scoring validation (starter pilot)** | Compare the app's scores with expert SLP ratings, acoustic measures, and parents' correct/incorrect judgments during home practice, across children with a range of speech motor skill. This replicates in English the human-versus-machine design of Getman et al. (2022), with several SLP raters instead of one and the same metrics (accuracy, unweighted average recall, mean absolute error), and adds a parent comparison. Using several raters matters because expert listening has limits of its own (Meyer & Munson, 2021; Allison et al., 2025), and combining expert listening with instrumental measures is recommended (Namasivayam et al., 2025). |
| **2. Home practice dose (existing data)** | Test whether the amount, spacing, and accuracy of home practice predict gains, and how closely families complete SLP-assigned practice, using de-identified data under a data use agreement. |
| **Longer-term studies** | Once scoring is validated: randomized comparisons of feedback type and frequency within the app, and prospective outcome studies with a comparison group and independent, functional measures such as percentage of consonants correct (Shriberg et al., 1997) and the Intelligibility in Context Scale (McLeod et al., 2012), plus stimulability and self-monitoring. |
| **Research partner** | SpeechLP is actively seeking a university research partner; finding a good fit is the first step. The studies above are proposals to be finalized with that partner. |

### Why generalization must be measured directly

Gains on practiced items do not guarantee carryover. In a three-child home-delivered articulation study, transfer across tested settings exceeded transfer to untrained words (Costello & Bosler, 1976). Self-monitoring has been studied as a way to support generalization, although its relationship with conversational gains is variable (Koegel et al., 1986; Shriberg & Kwiatkowski, 1990). A meta-analysis of 12 treatment studies found that treating more complex sounds also improved untreated, simpler sounds (Maggu et al., 2021), but this does not establish transfer from every form of app practice. Outcome studies will therefore measure untrained words and connected speech rather than relying on in-app scores.

### Implementation and Instructional Support

SpeechLP runs on iPhone and iPad and in a web browser, so children can practice at home, in the classroom, or in the therapy room. Children can start right away with a short try round, the age-based screener is built into the first levels, and the guided learning path tells the child what to do next, minimizing setup for adults. The iOS app requires iOS 17 or later and a one-time download of about 600 MB; a headset microphone and a quiet space improve speech recognition.




**The SLP workflow.** SpeechTherapist.app gives speech-language pathologists one place to manage SpeechLP practice for their caseload:

- **Invite and assign:** invite a caregiver with a connection code, then assign target sounds, word positions, and blends.
- **Monitor:** track games played, words practiced, and accuracy overall and by sound.
- **Plan:** use that data to decide what to target next and spend session time on new skills.

**Privacy.** SpeechLP complies with COPPA, FERPA, HIPAA, and PIPEDA. Speech recognition runs on the device where supported. Apple's speech recognition, used as a backup checker, may process audio on the device or on Apple's servers, subject to Apple's terms of service. SpeechLP never receives any audio. Transcribed text is stored as depersonalized data: all direct and indirect personal identifiers are removed or masked so it can no longer be used to identify a specific individual. An optional "Record a run" feature saves the child's voice on the family's own device.

Together, SpeechLP and SpeechTherapist.app give SLPs visibility into home practice, so session time can focus on new skills and families can reinforce the same targets.

## Conclusion

SpeechLP brings together several research-supported ingredients of speech sound intervention—age-based selection of target sounds, expert modeling, high-dose practice across many words, motivating game-based practice, and family involvement—in a tool children can use between SLP sessions. An initial descriptive study found higher automated accuracy scores on matched sounds in a second month of use among returning users, and the logic model and planned evaluation provide a clear rationale for its expected impact and a path to building stronger evidence of effectiveness.

# SpeechLP Logic Model

## PROBLEM STATEMENT

Speech sound disorders are among the most common communication difficulties of childhood, and untreated or persistent errors can affect a child's intelligibility, literacy, confidence, and participation in school. Progress depends on frequent, accurate practice, but children typically see an SLP for limited time each week, SLP caseloads are large, and many families lack access to services or do not know how to practice at home. Practice materials are often repetitive and fail to hold children's attention. SpeechLP helps close this gap with an engaging, SLP-guided practice app that identifies each child's tricky sounds and delivers personalized, high-dose practice at home or school, with progress reports that keep parents, teachers, and SLPs connected.

## RESOURCES

- Age-based screeners for ages 3–9, about 15 words each, sampling initial, medial, and final positions
- Short video lessons with models by Mari Gottlieb, M.A., CCC-SLP, and Alissa Fox, M.S., CCC-SLP; cueing written by Alissa and delivered by Mari
- Library of 1,500+ target words organized by sound and position
- Word-level speech practice games (e.g., Beary Hungry, Baby Talk, Chicken Little, Pilot Penguin)
- Guided learning path with levels and rewards
- Progress reports for parents, teachers, and SLPs
- SpeechTherapist.app, the SLP companion for assigning practice and monitoring progress
- iPhone/iPad app (iOS 17+) and web version, used with a microphone

## STRATEGIES AND ACTIVITIES

- Child starts with a short try round and completes the age-based screener built into the early levels; results identify target sounds.
- Child watches SLP video lessons modeling the target sound and practices saying it.
- Before each game, the child can preview and listen to the target words.
- Each practice card plays a spoken model of the word, which the child can replay before producing it.
- Child produces many different target words in short, game-based sessions.
- Child chooses sounds to play with and advances along a map, earning rewards.
- Parents, teachers, and SLPs review reports and reinforce the same targets.
- SLPs assign target sounds and practice in SpeechTherapist.app, then review each student's progress to plan sessions.

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## OUTPUTS

- Each child's tricky sounds are identified relative to age expectations.
- Children view expert SLP models of each target sound.
- Children complete a high number of target productions per session across many different words.
- Children complete practice sessions regularly at home and/or school.
- Parents, teachers, and SLPs receive regular progress reports.
- SLPs have current data on each student's assigned practice and progress across their caseload.

## SHORT-TERM AND INTERMEDIATE OUTCOMES

Improved accuracy of target sounds in practiced words. Increased awareness of how target sounds are produced. Greater willingness to practice and increased practice frequency. Earlier identification and referral of children who may need an SLP evaluation.

As children progress through SpeechLP, they are expected to show improved accuracy of target sounds in connected speech and conversation, generalization to untrained words, improved overall intelligibility, increased confidence when speaking, stronger home–school–SLP collaboration, and more efficient use of SLP session time.

## LONG-TERM OUTCOMES AND IMPACTS

- Age-appropriate speech sound production and intelligibility
- Fewer residual speech errors persisting into later grades
- Shorter time to dismissal from speech services
- Reduced risk of related literacy difficulties
- Greater confidence and participation in classroom and social communication
- Increased family capacity to support communication development

## ASSUMPTIONS

- An adult (parent, teacher, or SLP) is available to support the child as needed.
- Access to an iPhone, iPad, or computer with a working microphone and a reasonably quiet practice space.
- Access to the internet.
- Children with suspected speech sound disorders are also evaluated and served by a licensed SLP where available.

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Research partnership inquiries: alissa@speechlp.com

**The SpeechLP team**
Mari Gottlieb, M.A., CCC-SLP · Cofounder & Co-CEO
Alissa Fox, M.S., CCC-SLP · Cofounder & Co-CEO
Shail Silver · Cofounder & Chairman
Jabali Ndungu · Head of Engineering
Matthew Siu · Software Engineer
Cindy Cheng · UX/UI Designer